

SEUNGMIN HAN

Systems Architect · Capital Markets · Low-Latency Trading Systems

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PROFESSIONAL EXPERIENCE

Korea Investment Securities (한국투자증권) — Investment Engineering Dept. · Seoul Sep 2024 – Present

Market-Making / Liquidity-Provision (MM/LP) Systems Developer

- **Solely re-architected and fully rebuilt** the desk's MM/LP system covering **~5,500 instruments** (futures, options, ELW, ETN) — full-stack across the entire stack: from the backtesting database to a low-latency C++/C# back-end and a C# / DevExpress front-end.
- **Ported** the hot path from Windows/C# to Linux/C++ and applied CPU pinning, a busy-spin event loop, lock-free queues, and a pool allocator — **cutting tick-to-trade latency from 5 ms to ~25 μs (200×)**, perf-profiled; DPDK-based kernel-bypass integration in progress.
- **Specced and validated** a network switch replacement and NIC kernel-parameter tuning with the infrastructure team, further cutting network-layer latency.
- **Implemented** quoting logic for options, basket, and single-stock futures/options arbitrage strategies with the quant-trading desk — including microprice-based fair-value pricing and real-time quote adjustment.

University of Pennsylvania · Philadelphia, PA Jan 2022 – May 2024

Teaching Assistant → Head Teaching Assistant — CIS 5480 Operating Systems (graduate, ~250 students)

- Added **fuzz-style automatic edge-case generation** to the Python auto-grader, expanding test coverage across students' C operating-system implementations.
- Designed and taught a Linux-style OS capstone module (process scheduler and file system in C).
- Ran weekly code reviews of low-level C systems code (scheduling, virtual memory, syscalls) for graduate-level OS assignments.

Supergate · Seoul May–Aug 2022 & 2023

Software Engineer Intern — Applied Cryptography

- Researched fully homomorphic encryption (FHE) applications and built a **searchable FHE database library in C++** — supporting exact-match queries over encrypted data via a C++ · MongoDB back-end.
- Implemented two new CRUD-capable **Private Query protocols**, cutting **query-result size ~50%** on average.
- **Filed two Korean patents** (10-2023-0115045, 10-2023-0115051): "Encrypted Reverse Index" and "PDQ Protocol over Finite Extension Fields".
- Cross-compiled the database server and dependencies onto a **Xilinx Zynq FPGA** board.

Karrot (Daangn) · Korea Jun–Sep 2021

Global Technical Product Management Intern

- Drove **150% MAU growth** for the global marketplace app "Karrot" in the Toronto market.
- Built a real-time in-app user-behavior dashboard with BigQuery and Data Studio.
- Proposed a new consumer-penetration strategy via PMF (product–market–fit) analysis from prospective-user surveys.

Unbounded Research in Biomedical Systems Lab · USA May–Sep 2019

ML Research Assistant

- Implemented an ensemble of **three Relief-based ML algorithms** in scikit-learn, improving feature-selection / ranking speed by **150%** on large biomedical datasets.

SELECTED PROJECTS

Fully Secure Online Voting Platform using FHE — C++ · HElib · Senior Design, "Most Technically Challenging" award

- Led an end-to-end encrypted election system in which ballots are tallied entirely under encryption.

EDUCATION

University of Pennsylvania — School of Engineering & Applied Science · Philadelphia, PA

- **M.S.E., Computer & Information Science** — May 2024. Thesis: *Efficient Private Query Protocol using Fully Homomorphic Encryption*.
- **B.S.E., Computer & Information Science**, Minor in Mathematics — 2023. GPA 3.64/4.00; Dean's List (2021–2023); CIS Senior Design "Most Technically Challenging" award.

PATENTS (FILED)

- "Encrypted Reverse Index" — Korean patent application **10-2023-0115045**.
- "PDQ Protocol Using Finite Extension Fields" — Korean patent application **10-2023-0115051**.

EXPERTISE

Capital Markets & Trading Systems: market making / liquidity provision, index arbitrage, derivatives hedging, market data, exchange (KRX / FEP) connectivity, kernel-bypass networking, low-latency execution, pre-trade risk.

Cryptography & Security: fully homomorphic encryption, privacy-preserving protocols, applied security, crypto & market-data / PnL systems.

Engineering: C++20, C# / .NET 8, Python, TypeScript / React, SQL · Linux & Windows, DevExpress / WinForms · concurrency & lock-free systems, distributed & real-time architecture · FastAPI / HTMX, PostgreSQL / TimescaleDB · machine learning.

Architecture, Product & Delivery: system architecture, product management, data-driven growth, technical-to-business translation, cross-functional leadership.

Languages: Korean (native), English (fluent — U.S. educated).